
Matrices

A matrix is an array of numbers with m rows and n columns. Each row is a vector of length n and each column is a vector of length m .

When the number of rows equals the number of columns it is called a square matrix. When matrices are used for vector dynamics, they will be square matrices.

Dot product

Matrix multiplication is based on the dot product of vectors.

The dot product of two vectors is calculated by multiplying each component together, and then adding them all together.

$$\text{Out}[\text{ }]= \begin{pmatrix} 2 & 4 & 0 & -1 \end{pmatrix} \cdot \begin{pmatrix} 3 \\ 1 \\ -1 \\ 0 \end{pmatrix} = 2(3) + 4(1) + 0(-1) + (-1)(0) = 6 + 4 + 0 + 0 = 10$$

One thing to notice is the dot product of two vectors only makes sense if they have the same length.

Let's write the first vector in a row and the second in a column, since we are getting to matrix multiplication, where the first vector is a row of the first matrix, and the second vector is a column of the second matrix.

Matrix Multiplication

Using the dot product, we can describe matrix multiplication of matrices $A = (a_{ij})$ and $B = (b_{ij})$ as a matrix $C = (c_{ij})$ where the entry in i -th row and j -th column is named c_{ij} and is equal to the dot product of the i -th row of A and the j -th column of B .

The dot product only makes sense if the vectors have the same length. So for matrix multiplication to make sense, the length of a row in A has to be the length of a column in B . In other words, the number of columns in A has to be the same as the number of rows in B .

Let's walk through an example where the second matrix is a column vector.

$$\text{Out}[\text{ }]= A = \begin{pmatrix} 2 & 1 & 3 \\ -1 & 0 & 1 \end{pmatrix}$$

$$\text{Out}[\text{ }]= B = \begin{pmatrix} 4 \\ 5 \\ 0 \end{pmatrix}$$

The number of columns of A is 3 and the number of rows of B is 3 so $A \cdot B$ makes sense for matrix multiplication.

$$\text{Out}[\text{ }]= \begin{pmatrix} 2 & 1 & 3 \\ -1 & 0 & 1 \end{pmatrix} \cdot \begin{pmatrix} 4 \\ 5 \\ 0 \end{pmatrix} = \begin{pmatrix} ? \\ ? \end{pmatrix}$$

Start with the entry in the first row and first column. We dot product the first row of A with the first column of B.

$$Out[*]= \begin{pmatrix} 2 & 1 & 3 \\ -1 & 0 & 1 \end{pmatrix} \cdot \begin{pmatrix} 4 \\ 5 \\ 0 \end{pmatrix} = \begin{pmatrix} 2(4) + (1)(5) + 3(0) \\ ? \end{pmatrix}$$

$$Out[*]= \begin{pmatrix} 2 & 1 & 3 \\ -1 & 0 & 1 \end{pmatrix} \cdot \begin{pmatrix} 4 \\ 5 \\ 0 \end{pmatrix} = \begin{pmatrix} 8 + 5 + 0 \\ ? \end{pmatrix}$$

$$Out[*]= \begin{pmatrix} 2 & 1 & 3 \\ -1 & 0 & 1 \end{pmatrix} \cdot \begin{pmatrix} 4 \\ 5 \\ 0 \end{pmatrix} = \begin{pmatrix} 13 \\ ? \end{pmatrix}$$

Next up is the entry in the second row and first column. We dot product the second row of A with the first column of B.

$$Out[*]= \begin{pmatrix} 2 & 1 & 3 \\ -1 & 0 & 1 \end{pmatrix} \cdot \begin{pmatrix} 4 \\ 5 \\ 0 \end{pmatrix} = \begin{pmatrix} 13 \\ (-1)(4) + (0)(5) + 1(0) \end{pmatrix}$$

$$Out[*]= \begin{pmatrix} 2 & 1 & 3 \\ -1 & 0 & 1 \end{pmatrix} \cdot \begin{pmatrix} 4 \\ 5 \\ 0 \end{pmatrix} = \begin{pmatrix} 13 \\ -4 + 0 + 0 \end{pmatrix}$$

$$Out[*]= \begin{pmatrix} 2 & 1 & 3 \\ -1 & 0 & 1 \end{pmatrix} \cdot \begin{pmatrix} 4 \\ 5 \\ 0 \end{pmatrix} = \begin{pmatrix} 13 \\ -4 \end{pmatrix}$$

That's the answer.

$$Out[*]= \begin{pmatrix} 2 & 1 & 3 \\ -1 & 0 & 1 \end{pmatrix} \cdot \begin{pmatrix} 4 \\ 5 \\ 0 \end{pmatrix} = \begin{pmatrix} 13 \\ -4 \end{pmatrix}$$

Let's do a two by two example in this level of detail.

$$Out[*]= \begin{pmatrix} 2 & 1 \\ -1 & 0 \end{pmatrix} \cdot \begin{pmatrix} 4 & 0 \\ 5 & 3 \end{pmatrix} = \begin{pmatrix} ? & ? \\ ? & ? \end{pmatrix}$$

Start with the entry in the first row and first column. We dot product the first row of A with the first column of B.

$$Out[*]= \begin{pmatrix} 2 & 1 \\ -1 & 0 \end{pmatrix} \cdot \begin{pmatrix} 4 & 0 \\ 5 & 3 \end{pmatrix} = \begin{pmatrix} 2(4) + (1)(5) & ? \\ ? & ? \end{pmatrix}$$

$$Out[*]= \begin{pmatrix} 2 & 1 \\ -1 & 0 \end{pmatrix} \cdot \begin{pmatrix} 4 & 0 \\ 5 & 3 \end{pmatrix} = \begin{pmatrix} 8 + 5 & ? \\ ? & ? \end{pmatrix}$$

$$Out[*]= \begin{pmatrix} 2 & 1 \\ -1 & 0 \end{pmatrix} \cdot \begin{pmatrix} 4 & 0 \\ 5 & 3 \end{pmatrix} = \begin{pmatrix} 13 & ? \\ ? & ? \end{pmatrix}$$

Next let's do row 2 column 1

$$Out[*]= \begin{pmatrix} 2 & 1 \\ -1 & 0 \end{pmatrix} \cdot \begin{pmatrix} 4 & 0 \\ 5 & 3 \end{pmatrix} = \begin{pmatrix} 13 & ? \\ (-1)(4) + (0)(5) & ? \end{pmatrix}$$

$$Out[*]= \begin{pmatrix} 2 & 1 \\ -1 & 0 \end{pmatrix} \cdot \begin{pmatrix} 4 & 0 \\ 5 & 3 \end{pmatrix} = \begin{pmatrix} 13 & ? \\ -4 + 0 & ? \end{pmatrix}$$

$$Out[*]= \begin{pmatrix} 2 & 1 \\ -1 & 0 \end{pmatrix} \cdot \begin{pmatrix} 4 & 0 \\ 5 & 3 \end{pmatrix} = \begin{pmatrix} 13 & ? \\ -4 & ? \end{pmatrix}$$

Next let's do row 1 column 2

$$\text{Out}[*]= \begin{pmatrix} 2 & 1 \\ -1 & 0 \end{pmatrix} \cdot \begin{pmatrix} 4 & 0 \\ 5 & 3 \end{pmatrix} = \begin{pmatrix} 13 & (2)(0) + (1)(3) \\ -4 & ? \end{pmatrix}$$

$$\text{Out}[*]= \begin{pmatrix} 2 & 1 \\ -1 & 0 \end{pmatrix} \cdot \begin{pmatrix} 4 & 0 \\ 5 & 3 \end{pmatrix} = \begin{pmatrix} 13 & 0 + 3 \\ -4 & ? \end{pmatrix}$$

$$\text{Out}[*]= \begin{pmatrix} 2 & 1 \\ -1 & 0 \end{pmatrix} \cdot \begin{pmatrix} 4 & 0 \\ 5 & 3 \end{pmatrix} = \begin{pmatrix} 13 & 3 \\ -4 & ? \end{pmatrix}$$

Finally do row 2 column 2

$$\text{Out}[*]= \begin{pmatrix} 2 & 1 \\ -1 & 0 \end{pmatrix} \cdot \begin{pmatrix} 4 & 0 \\ 5 & 3 \end{pmatrix} = \begin{pmatrix} 13 & 3 \\ -4 & (-1)(0) + (0)(3) \end{pmatrix}$$

$$\text{Out}[*]= \begin{pmatrix} 2 & 1 \\ -1 & 0 \end{pmatrix} \cdot \begin{pmatrix} 4 & 0 \\ 5 & 3 \end{pmatrix} = \begin{pmatrix} 13 & 3 \\ -4 & 0 + 0 \end{pmatrix}$$

$$\text{Out}[*]= \begin{pmatrix} 2 & 1 \\ -1 & 0 \end{pmatrix} \cdot \begin{pmatrix} 4 & 0 \\ 5 & 3 \end{pmatrix} = \begin{pmatrix} 13 & 3 \\ -4 & 0 \end{pmatrix}$$

Eigenvalues and Eigenvectors

For eigenvalues and eigenvectors, let's postpone how to calculate them, and just talk about what they are.

Let A be a square matrix, v a vector and λ a number (called a scalar).

The eigenvalue-eigenvector equation is

$$A \cdot v = \lambda v$$

on the left hand side it is matrix multiplication and on the right hand side it is scalar multiplication.

For most vectors this won't work. For most scalars λ (lambda) this won't work either.

When it does work out then we say that v is an eigenvector with eigenvalue λ , or we might say λ is an eigenvalue with eigenvector v , or we might say they form a pair.

Here is an example. If you work out the matrix multiplication you get this.

$$\text{Out}[*]= \begin{pmatrix} 4 & 1 \\ -3 & 0 \end{pmatrix} \cdot \begin{pmatrix} -1 \\ 1 \end{pmatrix} = \begin{pmatrix} -3 \\ 3 \end{pmatrix}$$

On the other hand, this is just 3 times the vector.

$$\text{Out}[*]= \begin{pmatrix} -3 \\ 3 \end{pmatrix} = 3 \cdot \begin{pmatrix} -1 \\ 1 \end{pmatrix}$$

So these are the same.

$$\text{Out}[*]= \begin{pmatrix} 4 & 1 \\ -3 & 0 \end{pmatrix} \cdot \begin{pmatrix} -1 \\ 1 \end{pmatrix} = 3 \cdot \begin{pmatrix} -1 \\ 1 \end{pmatrix}$$

$(-1, 1)$ is an eigenvector with eigenvalue 3 for this matrix.

What if $A \cdot v = v$? Then we say that v is an eigenvector with eigenvalue 1

It turns out this matrix has another eigenvector-eigenvalue pair with $\lambda = 1$

$$\text{Out}[*]= \begin{pmatrix} 4 & 1 \\ -3 & 0 \end{pmatrix} \cdot \begin{pmatrix} -1 \\ 3 \end{pmatrix} = \begin{pmatrix} -1 \\ 3 \end{pmatrix}$$

General Facts

A general fact is that the number of eigenvalues does not exceed the size of the square matrix, and usually the number of eigenvalues is the same as the size of the matrix. There is always at least one eigenvalue. Two eigenvectors can share the same eigenvalue.

A very useful fact is that if v is an eigenvector that $2v$ is an eigenvector, because matrix multiplication is linear.

$$A \cdot (2v) = 2(A \cdot v)$$

So if v is an eigenvector,

$$A \cdot (2v) = 2(A \cdot v) = 2(\lambda v) = \lambda(2v)$$

then so is $2v$. There was nothing special about 2, it could have been any number. The only restriction on the eigenvector is that it is not zero.

All of these extra solutions to the eigenvector equation don't count when they are just scalar multiples, but it is beyond the scope of this discussion to describe the general situation. Instead we will think of v and $2v$ as being the same when it comes to the eigenvector-eigenvalue situation.

Calculating the Eigenvalue knowing the Eigenvector

If you are given a matrix A and an eigenvector it is easy to find the eigenvalue.

Examine this again. Suppose you are told

$$\text{Out}[*]= \begin{pmatrix} 4 & 1 \\ -3 & 0 \end{pmatrix} \text{ has eigenvector } \begin{pmatrix} -1 \\ 1 \end{pmatrix}$$

Then to get the eigenvalue, do the matrix multiplication

$$\text{Out}[*]= \begin{pmatrix} 4 & 1 \\ -3 & 0 \end{pmatrix} \cdot \begin{pmatrix} -1 \\ 1 \end{pmatrix} = \begin{pmatrix} -3 \\ 3 \end{pmatrix}$$

then you can divide the answer by the given vector to find that the eigenvalue is 3.

Calculating the Eigenvector knowing the Eigenvalue

On the other hand, suppose you are told the eigenvalue and want the eigenvector.

$$\text{Out}[*]= \begin{pmatrix} 4 & 1 \\ -3 & 0 \end{pmatrix} \text{ has eigenvalue } 3$$

The first step is to calculate the matrix multiplication symbolically.

$$\text{Out}[8]= \begin{pmatrix} 4 & 1 \\ -3 & 0 \end{pmatrix} \cdot \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} 4x + y \\ -3x \end{pmatrix}$$

The next step is to calculate the scalar multiplication symbolically. Remember you were told the eigenvalue is 3.

$$\text{Out}[9]= 3 \cdot \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} 3x \\ 3y \end{pmatrix}$$

Equating these two symbolic vectors gives you

$$\text{Out}[9]= \begin{pmatrix} 4x + y \\ -3x \end{pmatrix} = \begin{pmatrix} 3x \\ 3y \end{pmatrix}$$

Which gives you two equations, which both boil down to $y = -x$

It will always turn out that the equations are redundant.

It might seem tricky to find the solution, until you remember that the eigenvectors are only defined up to scalar multiplication (see above).

All you need to do to find an eigenvector is to plug in a nonzero value for either x or y and get the other one.

Here the equation was $y = -x$, so if we pick $x = 1$ then $y = -1$ and our eigenvector is $(1, -1)$

Can you always pick $x = 1$? Almost always. The one time you can't is if the equation is $x = 0$ instead of $y = -x$. If it is $x = 0$ then you can instead choose $y = 1$ and your eigenvector would be $(0, 1)$

Calculating the Eigenvalues

The general technique is to calculate the eigenvalues and then use the above technique to find the eigenvectors.

If you know that A is a transfer matrix then you know that 1 is an eigenvalue. That is one way to find an eigenvalue, by knowing the matrix is a transfer matrix. Check it by adding its columns and seeing if the total equals 1 for each column.

The general case for finding eigenvalues is to use technology, but the 2×2 case can be done by hand with the quadratic formula.

$$\text{Out}[81]= -b c + a d - a \lambda - d \lambda + \lambda^2$$

The eigenvalues λ of the 2×2 matrix

$$\text{Out}[82]//\text{MatrixForm} = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$$

are the two solutions λ to $\lambda^2 + (-a - d)\lambda + (ad - bc) = 0$

For example, for this matrix $a = 1$, $b = -2$, $c = 1$, $d = 4$

$$\text{Out}[86]//\text{MatrixForm} = \begin{pmatrix} 1 & -2 \\ 1 & 4 \end{pmatrix}$$

we get $\lambda^2 + (-1 - 4)\lambda + ((1)(4) - (-2)(1)) = \lambda^2 - 5\lambda + 6 = 0$

Using the quadratic formula we get

$$\lambda = \frac{5 \pm \sqrt{5^2 - 4(1)(6)}}{2} = \frac{5 \pm \sqrt{25 - 24}}{2} = \frac{5 \pm 1}{2} = 4/2 \text{ or } 6/2 = 2 \text{ or } 3$$

Note that in this case you could solve by factoring, which is faster, but the quadratic formula guarantees you get the answer.

One final comment: the quadratic formula can give no real solutions, in which case there are no real eigenvalues. Instead the eigenvalues will be complex numbers.